

PRACTICAL LAB: JAVASCRIPT OPERATORS MASTERY

Course: DFK20013 Web Design Technology

Topic: 3.1.1 Variables, Data Types, and Operators

Duration: 60 – 90 Minutes

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1. Learning Outcomes

By the end of this lab, students should be able to:

- Implement **Arithmetic Operators** for calculations.
- Use **Assignment Operators** to update values efficiently. • Control application states using **Increment/Decrement**.
- Validate user access using **Relational** and **Logical** operators.

2. Practical Task: The All-in-One Operator Workshop

Create a file named lab_operators.html. Copy and complete the coding below.

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>DFK20013: JS Operators Workshop</title>
  <style>
    body { font-family: 'Segoe UI', sans-serif; max-width: 850px;
margin: 2rem auto; background-color: #f0f2f5; padding: 20px; color: #333;
}
    section { background: white; padding: 25px; border-radius: 12px;
margin-bottom: 25px; box-shadow: 0 4px 6px rgba(0,0,0,0.1); }
    h2 { color: #2c3e50; border-left: 5px solid #3498db; padding-left:
10px; margin-top: 0; }
    .output-box { background: #2c3e50; color: #27ae60; padding: 15px;
border-radius: 6px; font-family: 'Courier New', monospace; margin-top:
10px; }
```

```

        .short-note { background: #fff3cd; color: #856404; padding: 15px;
border-radius: 6px; margin-top: 15px; border-left: 5px solid #ffc107;
display: none; font-size: 0.9rem; }

        input { padding: 8px; width: 100px; border-radius: 4px; border:
1px solid #ccc; }

        button { padding: 8px 20px; background: #3498db; color: white;
border: none; border-radius: 4px; cursor: pointer; font-weight: bold; }
button:hover { background: #2980b9; }

        code { background: #eee; padding: 2px 4px; border-radius: 3px;
font-weight: bold; color: #d63384; }

</style>

```

```
</head>
```

```
<body>
```

```

        <header style="text-align: center; margin-bottom: 30px;">

            <h1>JavaScript Operators Mastery</h1>

            <p>Follow the steps and observe the Console and UI outputs.</p>
        </header>

```

```

        <section>

            <h2>Task 1: Basic Arithmetic (Check Console)</h2>

            <p>This task demonstrates background calculations using
<code>+</code>, <code>-</code>, <code>*</code>, <code>/</code>,
<code>**</code>, and <code>%</code>.</p>

            <button onclick="runTask1()">Run Console Logic</button>

        </section>

```

```

        <section>

            <h2>Task 2: POS System (Arithmetic & Assignment)</h2>

            <p>Price: RM 50.00. <strong>Promo:</strong> Buy > 4 units to get
RM 20.00 discount.</p>

            Quantity: <input type="number" id="qty" value="1">

            <button onclick="runTask2()">Calculate Total</button>

            <div id="outPOS" class="output-box">Final Total: RM 0.00</div>

            <div id="notePOS" class="short-note"></div>

        </section>

```

<pre> <section> <h2>Task 3: Stock Control (Increment & Decrement)</h2> </pre>	<pre> <p>Simulating inventory updates.</p> Stock Level: 10

 <button onclick="runTask3('inc')">Add (+)</button> <button onclick="runTask3('dec')" style="background:#6c757d;">Remove (-)</button> <div id="noteStock" class="short-note"></div> </section> </pre>
<pre> <section> <h2>Task 4: Gate Access (Relational & Logical)</h2> <p>Rules: Age 18+ OR (Age 13+ AND Parent present).</p> Age: <input type="number" id="age" placeholder="Age"> <label><input type="checkbox" id="parent"> With Parent?</label>

 <button onclick="runTask4()" style="background:#27ae60;">Verify Entry</button> <div id="outSecurity" class="output-box">Status: Waiting...</div> <div id="noteSecurity" class="short-note"></div> </section> </pre>	

Inside the <script> tag, type all the following code:

JAVASCRIPT:

1.	<pre> <script> // Helper to show explanations function showNote(id, text) { const noteDiv = document.getElementById(id); noteDiv.style.display = 'block'; noteDiv.innerHTML = text; } </pre>
----	--

2.	<pre> // --- TASK 1: ARITHMETIC (BASIC) --- function runTask1() { console.log("--- Task 1: Arithmetic Operations ---"); let students = 31; students = students + 1; // Addition students = students - 2; // Subtraction students = students * 2; // Multiplication students = students / 2; // Division let power = 10 ** 2; // Exponentiation (100) let remainder = 31 % 2; // Modulus (1 - Odd) console.log("Final Student Count: " + students); console.log("10 to the power of 2: " + power); console.log("Remainder of 31/2: " + remainder); alert("Check Browser Console (F12) for results!"); } </pre>
3.	<pre> // --- TASK 2: POS SYSTEM (ARITHMETIC & ASSIGNMENT) --- function runTask2() { let price = 50; let qty = parseInt(document.getElementById('qty').value); // a. Arithmetic Operator (*) let total = price * qty; let note = "Subtopic (a) Arithmetic: <code>*</code> multiplied " + price + " by " + qty + ".
"; // b. Assignment Operator (-=) if (qty > 4) { total -= 20; // Shorthand for total = total - 20 note += "Subtopic (b) Assignment: <code>-=</code> updated the total by subtracting the discount."; } document.getElementById('outPOS').textContent = "Final Total: RM " + total.toFixed(2); showNote('notePOS', note); } </pre>

4.

```
// --- TASK 3: INCREMENT & DECREMENT ---
let stock = 10;
function runTask3(type) {
    let note = "";
    if (type === 'inc') {
        stock++; // (c) Increment
        note = "<b>Subtopic (c) Increment:</b> <code>++</code> increased the stock by 1.";
    } else {
        stock--; // (c) Decrement
        note = "<b>Subtopic (c) Decrement:</b> <code>--</code> decreased the stock by 1.";
    }
    document.getElementById('stockVal').textContent = stock;
    showNote('noteStock', note);
}
```

5.

```
// --- TASK 4: RELATIONAL & LOGICAL ---
function runTask4() {
    let age = parseInt(document.getElementById('age').value);
    let hasParent = document.getElementById('parent').checked;
    let status = document.getElementById('outSecurity');

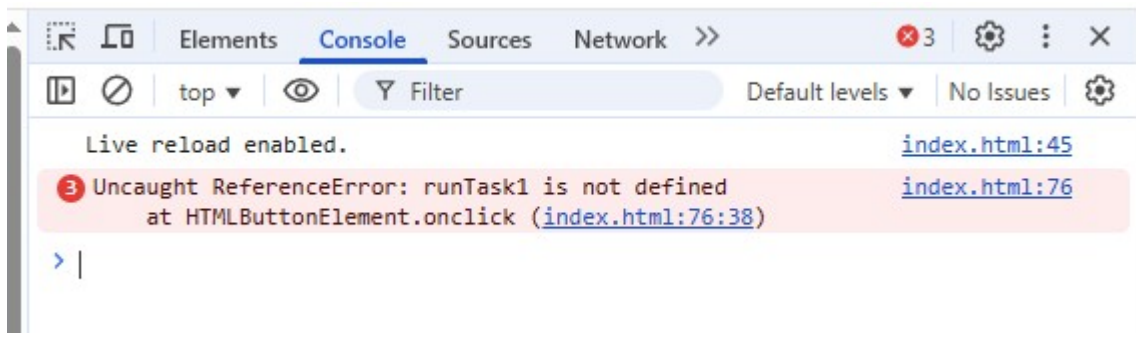
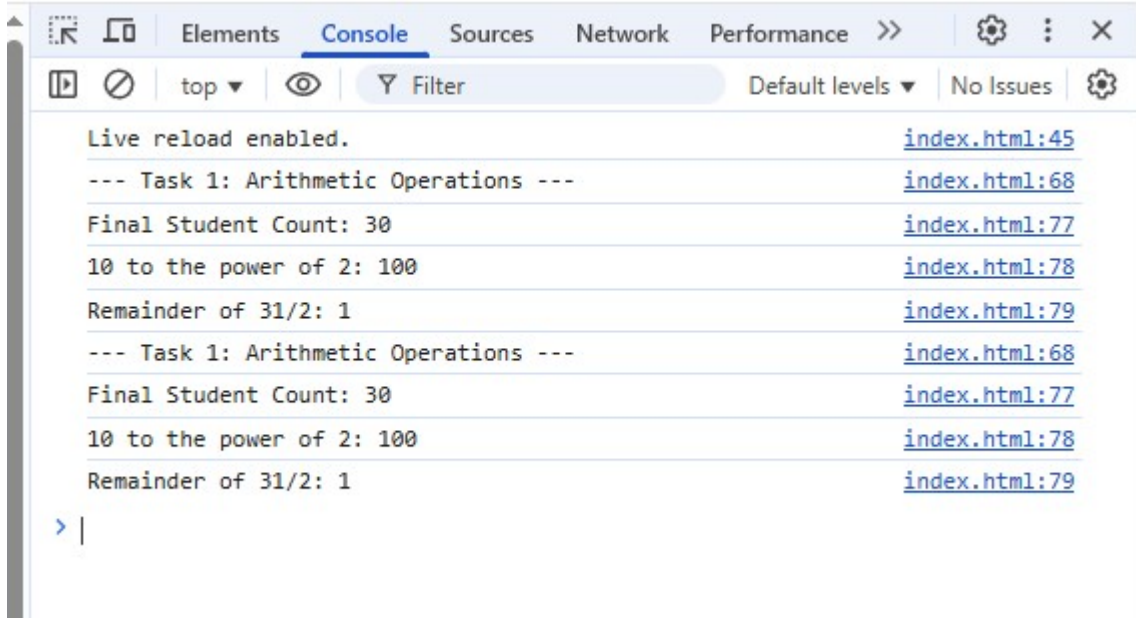
    if (isNaN(age)) return alert("Please enter a valid age.");

    // d. Relational: >=
    // e. Logical: || (OR), && (AND)
    let isEligible = (age >= 18) || (age >= 13 && hasParent);

    status.textContent = isEligible ? "Status: ACCESS GRANTED  
✔" : "Status: ACCESS DENIED ✖";
    status.style.color = isEligible ? "#27ae60" : "#ff4d4d";

    let note = "<b>Subtopic (d) Relational:</b> <code>>=</code> compared Age (" + age + ") to the limit.<br>";
    note += "<b>Subtopic (e) Logical:</b> Used <code>||</code> (OR) and <code>&&</code> (AND) to combine rules.";
    showNote('noteSecurity', note);
}
</script>
```

Capture a screenshot of the output from the previous steps. **Observe** the results and **explain** your understanding of the process.

1	Screenshot of the output 
	Explanation Nothing happens and the console doesn't work
2	Screenshot of the output 
	Explanation This task demonstrates how JavaScript handle basic mathematical calculations.
3.	Screenshot of the output

	Task 2: POS System (Arithmetic & Assignment) Price: RM 50.00. Promo: Buy > 4 units to get RM 20.00 discount. Quantity: 7854 <button>Calculate Total</button> Final Total: RM 392680.00 Subtopic (a) Arithmetic: * multiplied 50 by 7854. Subtopic (b) Assignment: -= updated the total by subtracting the discount.
	Explanation This section applies operators to a real-world Point of Sale (POS) scenario

4.	Screenshot of the output Task 3: Stock Control (Increment & Decrement) Simulating inventory updates. Stock Level: 11 <button>Add (+)</button> <button>Remove (-)</button> Subtopic (c) Decrement: -= decreased the stock by 1. Task 3: Stock Control (Increment & Decrement) Simulating inventory updates. Stock Level: 9 <button>Add (+)</button> <button>Remove (-)</button> Subtopic (c) Decrement: -= decreased the stock by 1.
	Explanation This task simulates inventory management using shorthand update operators

5. Screenshot of the output

Task 4: Gate Access (Relational & Logical)

Rules: Age 18+ OR (Age 13+ AND Parent present).

Age: ☒ With Parent?

Verify Entry

Status: Waiting...

Task 4: Gate Access (Relational & Logical)

Rules: Age 18+ OR (Age 13+ AND Parent present).

Age: ☐ With Parent?

Verify Entry

Status: Waiting...

Task 4: Gate Access (Relational & Logical)

Rules: Age 18+ OR (Age 13+ AND Parent present).

Age: ☒ With Parent?

Verify Entry

Status: ACCESS GRANTED ☒

Subtopic (d) Relational: `>=` compared Age (17) to the limit.

Subtopic (e) Logical: Used `||` (OR) and `&&` (AND) to combine rules.

Explanation

This task implements a security logic gate using comparison and conditional rules

3. Quick Check (Reflection)

Instruction: Answer the following questions based on your coding results.

1. **Modulus Utility:** Based on Task 1, if `students % 2` returns 0, is the number even or odd?
2. **Shorthand Power:** In Task 2, rewrite `total -= 20` using the standard assignment method.
3. **Logical Priority:** In Task 4, why did we put `(age >= 13 && hasParent)` in parentheses?
4. **Time Calculation:** (Extra Challenge) If a variable `totalMinutes = 130`, write the formula to find the remaining minutes after converting to hours using the **Modulus** operator.

ANSWER

1. **Modulus Utility:** If `students % 2` returns 0, the number is even. (A remainder of 0 means the number is perfectly divisible by 2).
2. **Shorthand Power:** The standard assignment method for `total -= 20` is:
`total = total - 20;`
3. **Logical Priority:** The parentheses are used to group the AND (&&) condition together as a single requirement. It ensures that the logic checks if the user is either 18+ OR (at least 13 with a parent). Without parentheses, the logic might evaluate in an unintended order due to operator precedence.
4. **Time Calculation:** To find the remaining minutes using the Modulus operator: let
`remainingMinutes = totalMinutes % 60;` (For 130 minutes, this would return 10, as $130 \div 60 = 2$ hours with 10 minutes remaining).